

INGREDION THAILAND — SPRINT 18K SPRAY DRYER PLANT (KALASIN)

REDUNDANT CISCO CORE SWITCHES & REDUNDANT SERVER INFRASTRUCTURE CONFIGURATION

DRAWING NO.
DWG-260911-NET-02

CISCO PROTOCOL
HSRP v2 + LACP

SERVER REDUNDANCY
FT VIEW SE CLUSTER

MCC TRUNK
802.1Q (VLAN 30,50,99)

SERVER ROOM LINK
DUAL 10G FIBER TRUNK

EXISTING SERVER ROOM (ZONE 1)

FIBER BACKBONE ENDPOINT

FDF-SR01	12-Core OM3/OS2 Fiber Optic Patch Panel (LC Duplex)
DC01: Primary Domain Controller	IP: 192.168.10.30 Active Directory / DNS / NTP Master HPE DL380 Gen10
DC02: Secondary Domain Controller	IP: 192.168.10.31 Active Directory Backup / Failover HPE DL380 Gen10
FT Historian SE 01 (Primary)	IP: 192.168.10.20 FactoryTalk Process Data Archive HPE DL380 Gen11
FT Historian SE 02 (Mirror)	IP: 192.168.10.21 OSIsoft PI Collective Mirror Node HPE DL380 Gen11
Server Room Online UPS Unit	IP: SNMP: 192.168.99.10 5kVA Redundant Dual-Power Source APC Smart-UPS RT

MAIN CONTROL PANEL (MCP-01) — CONTROLLOGIX CS1 - CS4

RACK 1 (CS1): MASTER CONTROLLER & DLR SUPERVISOR Slot 00: 1756-L950TPSXT Master Controller (50MB Process Safety) Slot 01: 1756-EN4TR (DLR Supervisor Node) — IP: 192.168.30.11 Slot 02: 1756-EN4TR (Control LAN SCADA Uplink) — IP: 192.168.30.20 Slots 03-12: Redesigned Balanced I/O (4 DI / 2 DO / 2 AI / 2 AO) Gateway VIP: 192.168.30.1
RACK 2 (CS2): Expansion Rack 1 (DLR Node 1) Slot 00: 1756-EN4TR (IP: 192.168.30.12) + I/O Modules & Spares
RACK 3 (CS3): Expansion Rack 2 (DLR Node 2) Slot 00: 1756-EN4TR (IP: 192.168.30.13) + I/O Modules & Spares
RACK 4 (CS4): Expansion Rack 3 (DLR Node 3) Slot 00: 1756-EN4TR (IP: 192.168.30.14) + I/O Modules & Spares

TECHNICAL ARCHITECTURE SPECIFICATIONS

- CISCO CORE SWITCH HIGH AVAILABILITY (HSRP v2):**
 - SW-CORE-01 is configured with Priority 110 (Active Gateway for all VLANs).
 - SW-CORE-02 is Standby (Priority 100). Automated failover time < 900 ms.
- SERVER REDUNDANCY & DUAL-HOMING:**
 - FT View SE Primary (SRV-01) & Secondary (SRV-02) have dual cross-connected NICs.
 - Private direct Cat6A Heartbeat link on NIC 3 ensures zero split-brain.
- 802.1Q TRUNK TO MCC ROOM:**
 - Carries VLAN 30 (Control), VLAN 50 (MCC Drives), and VLAN 99 (Management).
 - Connected to SW-MCC-01 Stratix 5700 switch feeding PowerFlex VFDs.
- FIBER OPTIC BACKBONE TO SERVER ROOM:**
 - Redundant 10G/1G multi-strand fiber connecting Core Switches to FDF-SR01.
 - Ensures continuous access to Historian, Batch Archives, and Active Directory.

OPERATION 1 ROOM — CENTRAL CONTROL & NETWORK CABINETS

SW-CORE-01 (CISCO CATALYST 9300)

★ HSRP ACTIVE (110)

Role: Primary Layer 3 Gateway & STP Root

Mgmt IP: 192.168.99.2 (VLAN 99)

VLAN SVI	SWITCH SVI IP	HSRP VIP (GATEWAY)	STATE
VLAN 10 (SCADA)	192.168.10.2	192.168.10.1	Active
VLAN 20 (HMI)	192.168.20.2	192.168.20.1	Active
VLAN 30 (PLC)	192.168.30.2	192.168.30.1	Active
VLAN 50 (MCC)	192.168.50.2	192.168.50.1	Active

ISL: LACP Po1 (20G)

SW-CORE-02 (CISCO CATALYST 9300)

HSRP STANDBY (100)

Role: Secondary Gateway & Hot Standby

IP: 192.168.99.3 (VLAN 99)

VLAN SVI	SWITCH SVI IP	HSRP VIP (GATEWAY)	STATE
VLAN 10 (SCADA)	192.168.10.3	192.168.10.1	Standby
VLAN 20 (HMI)	192.168.20.3	192.168.20.1	Standby
VLAN 30 (PLC)	192.168.30.3	192.168.30.1	Standby
VLAN 50 (MCC)	192.168.50.3	192.168.50.1	Standby

NIC1 → G11/0/1

NIC2 → SW2 G11/0/1 (Cross)

NIC2 → SW1 G11/0/2 (Cross)

NIC1 → G11/0/2

SRV-SCADA-01 (PRIMARY HMI SERVER)

● ACTIVE MASTER

Role: Primary FactoryTalk View SE Data & Alarm Server

Cluster IP: 192.168.10.10 (VLAN 10)

INTERFACE	TEAM / ROLE	CONNECTED TO	STATUS
NIC 1 (10G)	LACP Team A	SW-CORE-01 (G11/0/1)	Active
NIC 2 (10G)	LACP Team B	SW-CORE-02 (G11/0/1)	Standby Link
NIC 3 (Sync)	Heartbeat Private	Server 02 (10.10.10.1)	< 500ms
NIC 4 (iLO)	Hardware Mgmt	SW-MGMT-01 (VLAN99)	Ready

DEDICATED HEARTBEAT (10.10.10.x/30)

SRV-SCADA-02 (SECONDARY HMI SERVER)

● HOT STANDBY

Role: Secondary FactoryTalk View SE Data & Alarm Server

Cluster IP: 192.168.10.11 (VLAN 10)

INTERFACE	TEAM / ROLE	CONNECTED TO	STATUS
NIC 1 (10G)	LACP Team A	SW-CORE-02 (G11/0/2)	Active
NIC 2 (10G)	LACP Team B	SW-CORE-01 (G11/0/2)	Standby Link
NIC 3 (Sync)	Heartbeat Private	Server 01 (10.10.10.2)	< 500ms
NIC 4 (iLO)	Hardware Mgmt	SW-MGMT-01 (VLAN99)	Ready

CBL-UPL-01 (VLAN 30)

REDUNDANT CORE SWITCH & SERVERS

CONTROL ROOM CLIENTS & OVERVIEW DISPLAYS (VLAN 20)

Engineering Workstation (EWS) Studio 5000 / FT Admin / System Config	192.168.20.150
Operator Station 1 (Main Console) FT View SE Client / Batch Control	192.168.20.101
Operator Station 2 (Main Console) FT View SE Client / Dryer Control	192.168.20.102
Operator Station 3 (Remote Station 2) Remote Thin Client / Field Operation	192.168.20.103
55" Overhead Display: Process Overview Central High-Res Overview Monitor	ThinManager
55" Overhead Display: Trending & Alarms Process Curve & Realtime Trends	ThinManager

MCC ROOM — MOTOR CONTROL CENTER (ZONE 3)

802.1Q TRUNK: VLAN 30, 50, 99

SW-MCC-01: ALLEN-BRADLEY STRATIX 5700 (1783-BMS10CGN) Trunk Port 1: From SW-CORE-01 (G11/0/24) Trunk Port 2: From SW-CORE-02 (G11/0/24 - Standby) Gateway VIP: 192.168.50.1 IP: 192.168.50.10		
PowerFlex 755 (110kW) Supply Air Fan FN-60201 IP: 192.168.50.21 VLAN 50	PowerFlex 755 (132kW) Exhaust Air Fan FN-60202 IP: 192.168.50.22 VLAN 50	PowerFlex 525 (18.5kW) Slurry Transfer Pump PG-20201 IP: 192.168.50.31 VLAN 50
PowerFlex 525 (18.5kW) Slurry Transfer Pump PG-20211 IP: 192.168.50.32 VLAN 50	Power Meter 1408-BC3A Main Incoming 400V Metering IP: 192.168.50.50 VLAN 50	Agitator Starters (ME-20201/11) Slurry Tanks Agitator VFDs IP: 192.168.50.33 VLAN 50